

Experiment 3

Part 1: Hub

Aim

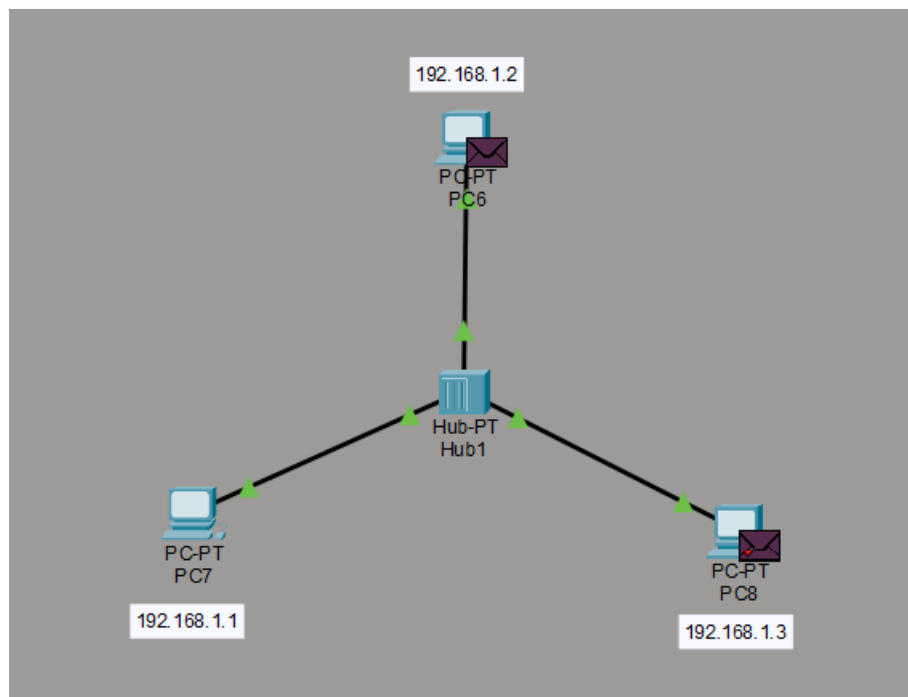
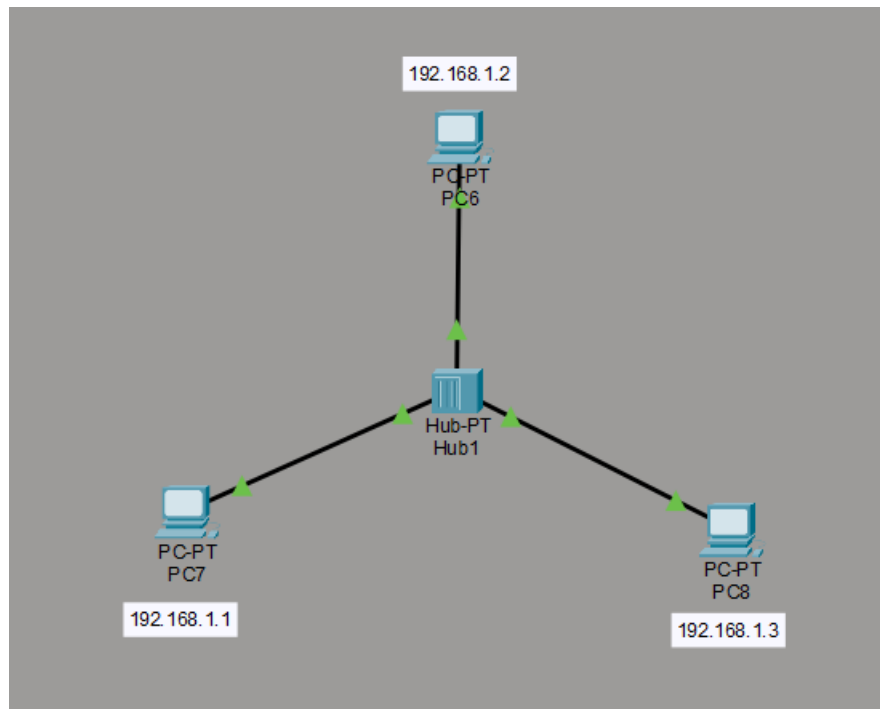
To understand the basic working of a Hub in Cisco Packet Tracer.

Description

A Hub is a Layer-1 networking device used to connect multiple computers in a Local Area Network (LAN). It does not maintain any address table and simply broadcasts every incoming data frame to all connected devices. Since all devices receive the data, only the intended destination processes it while the others discard it. Hubs are simple devices and are mainly used for learning basic networking concepts.

Procedure

1. Open Cisco Packet Tracer and create a new workspace.
2. 1. Hub section -> choose one hub -> click on hub and check for the ports
3. 2. Take 6 PC's and take straight through cable -> Connect all PC's with Hub with FastEthernet port
4. 3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.5 as Destination_ip



Observation

The Hub transmitted every frame to all connected devices. Communication was successful because the destination PC accepted the frame while the remaining devices ignored it.

Part 2: Switch

Aim

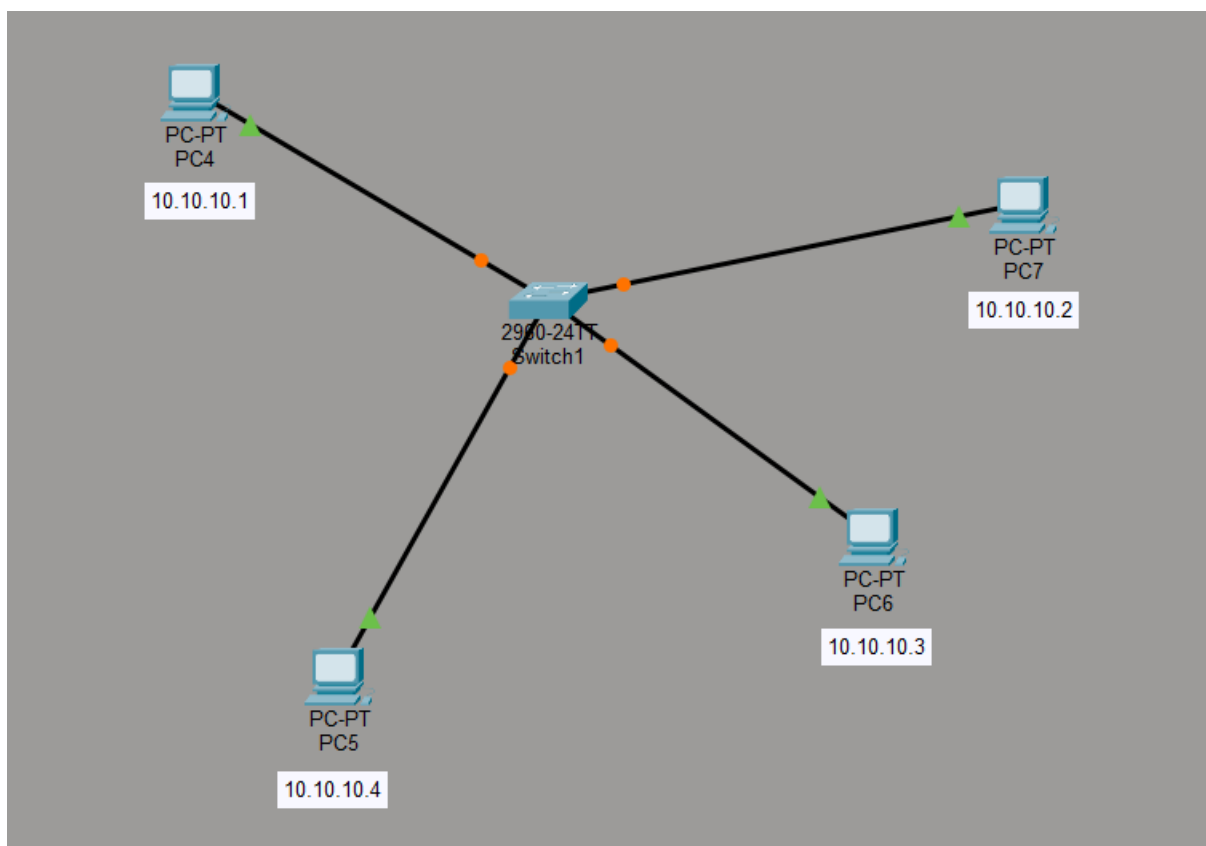
To understand the basic working of a Layer-2 Switch in Cisco Packet Tracer.

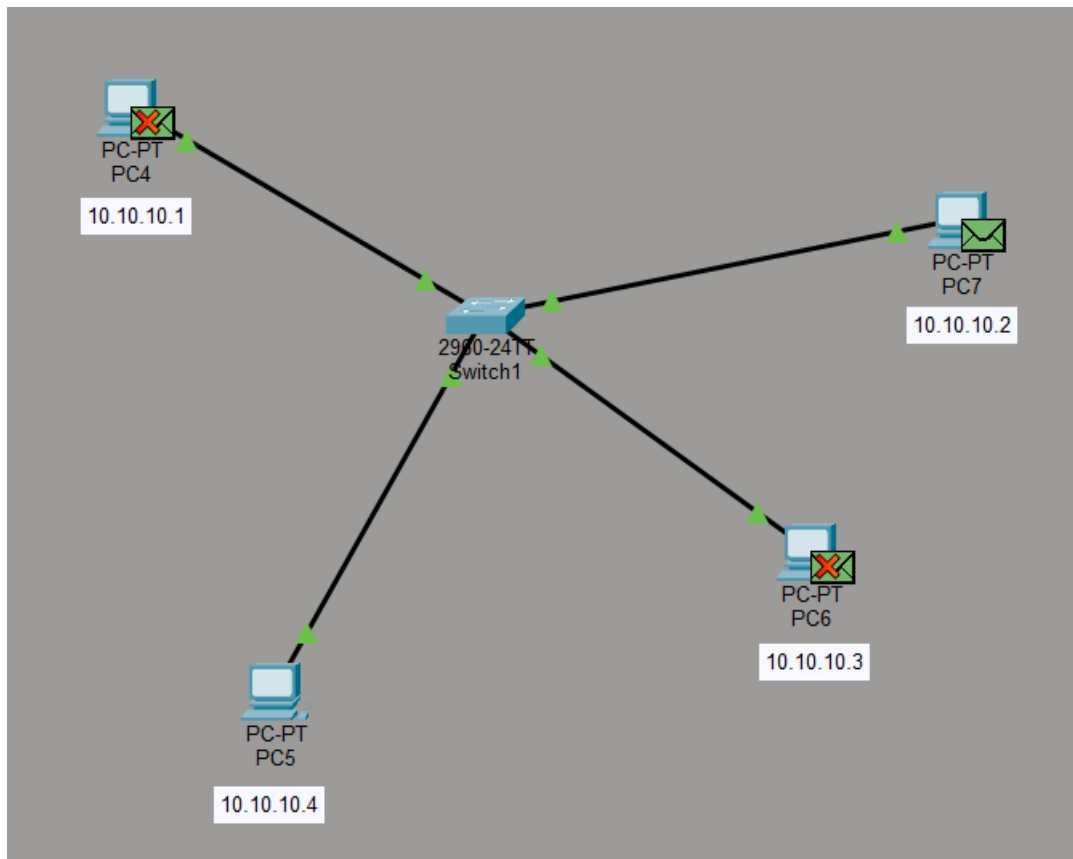
Description

A Switch is a Layer-2 networking device that forwards frames based on MAC addresses. It learns the MAC addresses of connected devices and stores them in a MAC address table. Unlike a Hub, it sends data only to the required destination, reducing unnecessary network traffic and improving communication efficiency.

Procedure

1. Place a Switch and two or more PCs in the Packet Tracer workspace.
2. Connect each PC to the Switch using Copper Straight-Through cables.
3. Configure valid IP addresses for all PCs in the same network.
4. Use the ping command to test connectivity.
5. Observe the simulation to see the Switch learn MAC addresses and forward frames only to the correct destination.





Observation

The Switch successfully learned the MAC addresses of connected devices and forwarded frames only to the intended destination instead of broadcasting them to every port.

Part 3: Router

Aim

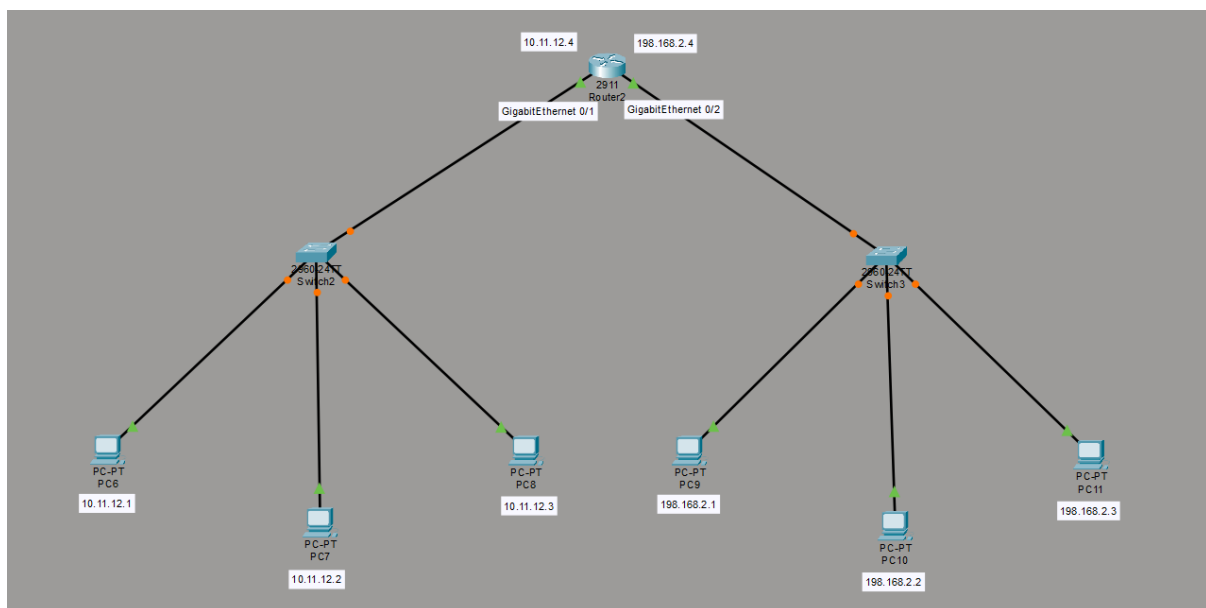
To understand the basic working of a Router in Cisco Packet Tracer.

Description

A Router is a Layer-3 networking device that connects different networks and forwards packets based on IP addresses. It determines the best path for data transmission and enables communication between devices located on different subnets. Routers are commonly used to interconnect LANs and provide access to external networks.

Procedure

1. Place a Router and two PCs in Cisco Packet Tracer.
2. Connect the PCs to different router interfaces using Copper Straight-Through cables.
3. Assign IP addresses to the PCs in different subnets.
4. Configure IP addresses on the Router interfaces using the Cisco IOS CLI and enable the interfaces using the no shutdown command.
5. Verify connectivity by using the ping command between the two PCs.
6. Observe packet forwarding in Simulation Mode.



Observation

The Router successfully forwarded packets between different networks after proper interface and IP configuration, allowing communication between devices on separate subnets.

Learning Outcomes

After completing this experiment, the student was able to:

1. Understand the basic working principles of a **Hub, Layer-2 Switch, and Router** in a computer network.
2. Differentiate between the functions of a Hub, Switch, and Router based on how they forward data.
3. Create simple network topologies using Hub, Switch, and Router in **Cisco Packet Tracer**.
4. Configure basic IP addressing for end devices and router interfaces.
5. Verify network connectivity using the **ping** command.